

CZ5101 Numerical Recipes, Midterm Test (closed book), 11:00am, Friday, 1 Oct 04

(with partial answers)

[Equal weights in marks for all questions]

1. Due to round off error, equivalent expressions mathematically need not yield the same numerical value. Which of the following expressions are preferred? Explain briefly why.
- (a) $a^2 - b^2$, (b) $(a+b)(a-b)$.

Ans: (b)

2. (a) Can we do LU decomposition for all 2×2 real square matrices (without pivoting)? Try to answer this question by looking at the example below:

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

(b) What if pivoting is allowed?

Ans: (a) No, (b) Yes.

3. (a) Consider interpolating 4 points $(x,y) = (0,1), (1,1), (2,1), (3,1)$ with a polynomial of degree 3 or less, without actually going through any calculation, what should be that polynomial?
- (b) Based on the result in (a), show that $\sum_{i=1}^N l_i(x) = 1$ for any x and N , where $l_i(x)$ is the Lagrange interpolation polynomials of order $N-1$ in the Lagrange's formula $f(x) = \sum_{i=1}^N l_i(x)y_i$. You don't need the specific form of $l_i(x)$ to prove this.

Ans: (a) $f(x) = 1$,

(b) Fit $f(x)$ to N points of $f(x_i)=1$, then $f(x) = 1$ by (a), and $y_i = 1$ for all i .

4. What is the accuracy of the rectangular integration rule (equally spaced abscissas):

$$\int_{x_0}^{x_N} f(x)dx = \frac{(x_N - x_0)}{N} \sum_{i=0}^{N-1} f_i + O\left(\frac{1}{N^k}\right)?$$

I.e., find the power k in the error $O(1/N^k)$.

Ans: $k=1$

5. If the sequence $\{5, 2, 7, 8, 9, 0, 4\}$ is given to the heap sort, what does the heap look like when the heap is just built (e.g., give the result of heap building pass where the top (root) gets the largest value).

Ans: $[9, 8, 7, 5, 2, 0, 4]$

6. Explain what are (a) linear convergence, and (b) quadratic convergence, in root finding.

Ans: Let ε_n be error in n -th iteration, then (a) linear convergence $\varepsilon_{n+1} = c \varepsilon_n$, (b) quadratic convergence $\varepsilon_{n+1} = c \varepsilon_n^2$.